An Analysis of United States Traffic Accidents

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Abstract

Using the US Accidents dataset—an archive of over 7 million geographically coded crash records—we investigated how temporal, spatial, and meteorological factors influence crash severity across the continental United States. While a range of methods were explored, our final analysis centered on three methods: ordinary least squares to assess nonlinearity, a 300-tree random forest for predictive modeling, and a two-way ANOVA to quantify interaction effects. Additionally, k-means clustering was utilized for interpretability. We find that crash severity increases substantially when poor visibility conditions occur at night. Two-way ANOVA revealed a statistically significant interaction between lighting and weather, with snow or ice during nighttime elevating average severity by 0.26 on a four-level scale. The random forest model explained 37% of out-of-sample variance, identifying visibility, time of day, and precipitation as the most influential features. K-means clustering over weather and temporal attributes isolated high-risk conditions—such as cold, low-visibility nights—with severe injury rates of 21.8%, compared to just 11.8% in clear daytime scenarios. These findings suggest that real-time visibility alerts, improved nighttime lighting, and weather-responsive traffic interventions could meaningfully reduce crash severity. Limitations include geographic bias toward urban roads, underreporting of driver-level data, and absence of contextual behavioral variables. Future research may address these concerns of further explore embedding the road map in a robust model or developing more complex and explanatory statistical models.

1 Dataset

The dataset used in this study is called *US Accidents (2016–2023)* and was created by **Sobhan Moosavi**, a researcher at The Ohio State University. The dataset was designed to support large-scale traffic accident prediction and analysis across the contiguous United States. The original dataset was introduced in 2019 and described in the paper *A Countrywide Traffic Accident Dataset* Moosavi et al. [2019b], and it has since been updated and maintained on Kaggle Moosavi [2023], where we accessed the version containing accident data from February 2016 to early 2023.

The dataset was created through a multi-stage process involving real-time streaming data collection from two commercial APIs—*MapQuest Traffic API* and *Microsoft Bing Maps Traffic API*. These APIs aggregate traffic incident data from sources such as state departments of transportation, police reports, traffic cameras, and road sensors. The team pulled data every 90 seconds during the day and every 150 seconds at night, collecting over 2.27 million reports. Duplicate entries between the two sources were removed using spatial and temporal thresholds, resulting in approximately 2.25 million unique accidents from 2016 to 2019, later expanded to 2023 for the Kaggle version (7.7 million unique accidents).

Each accident record includes up to 45 attributes, spanning multiple categories:

- **Traffic Attributes**: severity, time, GPS coordinates, distance, and natural language description
- **Address Attributes**: city, county, state, zip code
- **Weather Attributes**: temperature, humidity, precipitation, wind speed, visibility, and weather condition
- **Point-of-Interest (POI) Tags**: sourced from OpenStreetMap, such as traffic signals, stop signs, railway crossings, etc.
- **Period-of-Day Labels**: determined from accident start times using daylight phase APIs (e.g., civil twilight)

Weather data was retrieved from 1,977 weather stations using the *Weather Underground API*, with each accident matched to the closest station and timestamp. Reverse geocoding was performed using *Nominatim*, and POI features were linked based on spatial proximity optimized by Jaccard similarity, using regex-derived labels from accident descriptions as weak supervision.

The scale and richness of this dataset make it one of the largest and most robust open-access traffic accident datasets available. However, it is important to note that the dataset depends on real-time traffic reporting and may be biased toward more severe or urban incidents. Crashes in rural areas or those not reported through the APIs are likely underrepresented. Additionally, the dataset includes only incidents within the contiguous United States, excluding Alaska and Hawaii.

We downloaded the dataset as a `.csv` file from Kaggle in May 2025. It is hosted in the Kaggle Datasets library and is identified with DOI 10.34740/KAGGLE/DS/199387.

2 Research Question

The primary goal of this paper is to investigate if, and if so how well, environmental and temporal factors—such as weather conditions, lighting, and time of day—can predict the severity of U.S. roadway crashes. Severity, as defined by the dataset, is a four-level qualitative scale ranging from 1 to 4. This prompts several natural follow-up questions which our analysis seeks to address: Which individual or combined features most strongly correlate with increased severity? How does crash severity vary under specific weather–lighting interactions, such as nighttime precipitation or low visibility conditions? And can we identify interpretable scenarios or contexts in which severity risk meaningfully increases? Rather than attempting to fully explain severity, which is known to be highly stochastic, our goal is to isolate and quantify the most influential conditions and evaluate how these insights can support safety interventions.

2.1 Relevance

This research question is relevant and important to explore as US traffic accidents are a leading cause of death in the U.S. The National Highway Traffic Safety Administration estimated that in 2022, 42,514 deaths and $340 billion in comprehensive costs were caused by traffic accidents [National Highway Traffic Safety Administration, 2023]. These unfortunate fatalities may be preventable, or at the very least curtailed, however, it is not widely clear why severe accidents occur. Additionally, climate change and other factors may be increasing the frequency of extreme weather events which

may impact crash severity. Exploration into the factors that best explain severity will allow law makers, politicians, and academics to better understand the situation on a fundamental level; legislation or improved research may directly result in safer roadways. Pragmatically, understanding when and where severe crashes occur might yield safer, location-aware warnings in navigation services, guide government transportation entities in infrastructure design, and prompt reallocation of political spending in high-risk areas. Furthermore, by potentially applying statistical techniques such as ANOVA and PCA, we can isolate which conditions most strongly contribute to crash severity and uncover latent patterns in how environmental and roadway features interact.

2.2 Related Work

Nationwide accident risk research is still a relatively unexplored space as public data that scale has only become available within the last five years. The descriptive analysis by the publishers of the dataset found strong intra-day and weekday cycles of traffic accident frequency. Additionally, they found that low visibility and precipitation are associated with higher severity levels. However, they did not build predictive models that may be useful in a broader context Moosavi et al. [2019a]. A follow up study by the same group introduced a Deep Accident Prediction (DAP), a recurrent dense neural network that was fed by by sparse signals like traffic events, real time weather, and spatial points of interest. They trained their model on five large metros and achieved a AUROC (metric related to ratio of true positive against false positive rate) of 0.88 for 30 minute accident forecasting. This outperformed benchmark gradient boosting and classical logistic regression methods Moosavi et al. [2019c]. The drawback with this method was that it required very dense data feeds and the black box model limited external insights to legislators and policymakers.

Later, in 2022, Yan and Shen [2022] evaluated a random forest classier on a 2016 -2021 slice of the same US accidents dataset. After removing leakage features (ending lat./long., etc.), they achieved a $76\% - 80\%$ accuracy and identified wind speed, pressure, and visibility as the most explanatory predictors of severity. Their results were further evaluated and were found to be robust and were in line with earlier findings. In 2023, a paper provided a deep learning perspective that embedded accidents on a road graph. This explicitly modeled the spatial element of the dataset and was a novel approach. In particular, they constructed a 9 million record multi-state dataset that extended the US Accidents dataset with OpenStreetMap topology and traffic counts. The best model they developed was a multitask "GraphSAGE" variant (neural network) that predicted whether an accident would occur on a road segment. This model achieved an AUROC of 0.87 across 18 states Nippani et al. [2024]. This particular approach emphasized the importance of graph structure and neighboring crash history. However, this method still sacrificed transparency and an interpretable insight to the dataset.

3 Research Methods

This project uses the U.S. Accidents dataset (2016–2023), which contains over 7 million traffic crash records collected from various state and local agencies. It offers timestamped, geocoded observations with environmental, structural, and temporal metadata for each accident — including severity ratings, weather conditions, visibility, road infrastructure flags (e.g., traffic signal, bump), and time of day. Because our central research question focuses on how well environmental and temporal factors can predict severity, this dataset is uniquely well-suited. It offers high volume, national scale, and relevant feature coverage with sufficient variation and granularity for meaningful statistical exploration.

To explore this question rigorously and transparently, we employed a complementary set of analytical methods, each chosen to address a different dimension of the research problem. We began with ordinary least squares regression as a baseline model to assess linearity and establish a reference point for explanatory variables. To capture nonlinear relationships and feature interactions, we implemented a 300-tree random forest, which also enabled ranking of variable importance. We then used two-way ANOVA on grouped categorical variables to statistically validate and quantify key interaction effects—particularly between lighting and weather. Finally, to translate these statistical findings into insight, we applied k-means clustering on weather and temporal features to uncover interpretable environmental crash scenarios. Additional techniques such as principal component analysis (PCA), logistic regression and one-way ANOVA were used in exploratory stages to benchmark or support our core findings.

3.1 Justification for Methods and Alternative Techniques

We considered the following analytic techniques, ultimately emphasizing Random Forest, two-way ANOVA, and clustering based on their performance and explanatory value:

Technique	**Rationale**	**Attempted?**	**Yielded Insight?**	**Novel vs. Prior Work**
OLS Regression	Linear benchmark; shows how much structure a straight line misses.	Yes	Confirmed strong non-linearity ($R^2 \approx 0.06$).	Classical baseline in transport.
Random Forest	Captures non-linear interactions; supports permutation importance.	Yes	6× lift over OLS ($R^2 = 0.37$); identifies visibility-lighting pair.	Extends 2022 RF work to national scale with additional insight.
Two-Way ANOVA	Formal test of visibility × lighting interaction.	Yes	Both main and interaction effects highly significant ($p < 10^{-9}$).	Interpretability step rarely included in prior large-scale papers.
K-Means Clustering	Converts statistical interaction into nine interpretable "risk scenarios."	Yes	Flags a cold, low-visibility night cluster with $\approx$ 1.9× injury risk.	Absent in prior work.
Gradient Boosting (GBM)	Strong non-linear benchmark in Moosavi (2019).	No	-	Used only in earlier DAP benchmarks; not revisited here.
Deep Accident Prediction (DAP)	Recurrent dense NN fed by traffic + POI streams.	No	-	Prior work (Moosavi 2019); accurate but black-box and data-hungry.
GraphSAGE GNN	Embeds crashes on a road-network graph to exploit spatial context.	No	-	Prior work (Nippani 2023); novel but opaque; not reproduced here.

We began our analysis with ordinary least squares (OLS) regression to establish a linear benchmark for modeling crash severity. Using 24 predictors—including continuous weather features, lighting categories, and road geometry flags—the model achieved a holdout R^2 of just 0.06. The residuals displayed a flared shape, confirming that key relationships were nonlinear in structure. We implemented a 300-tree random forest to capture these nonlinearities, using five-fold cross-validation and class reweighting. This model improved predictive power to $R^2 = 0.369$, and permutation importance revealed that four features—visibility distance, hour of day, grouped weather conditions, and a binary night flag—contributed 58% of of the model's explanatory capacity. Partial dependence plots illustrated the interaction between lighting and visibility. Two-way ANOVA corroborated the significance of lighting–weather interactions on severity outcomes. Nine interpretable crash profiles emerged, with some high-risk clusters showing nearly double the severity rate of baseline conditions. Finally, to translate these patterns into interpretable real-world contexts, we applied k-means clustering on six environmental and temporal variables. Exploratory methods such as PCA and one-way ANOVA were not central to our final findings.

3.2 Implementation Overview

Our implementation followed four stages, with each method addressing limitations from the previous. All analysis was conducted in `R` using cleaned data, excluded missing values in key features, to evaluate complete weather-hour entries. Sampling, where needed for scalability, was done on subsets with complete weather-hour data.

Exploratory Data Analysis:

- Generated a hexbin plot of accident frequency across the continental U.S.
- Identified high-density regions along the East Coast, West Coast, and major cities such as New York, Los Angeles, Chicago, and Houston.
- Established that accident frequency alone does not imply severity.

Ordinary Least Squares (OLS):

- Modeled the four-level severity score using 24 predictors, including continuous weather, lighting categories, and road geometry flags.
- Achieved a holdout $R^2 = 0.060$, with residuals fanning strongly, indicating significant nonlinearity.
- Served primarily to demonstrate the inadequacy of linear assumptions.

Random Forest:

- Trained a 300-tree model with five-fold cross-validation and class reweighting.
- Improved predictive performance to $R^2 = 0.369$—a $6\times$ gain over the OLS benchmark.
- Identified top predictors: visibility distance, hour of day, grouped weather conditions, and a night flag, which together accounted for 58% of importance.

Partial Dependence Plot:

- Visualized the interaction between visibility and lighting.

- Showed steep risk increase in low-visibility nighttime scenarios.

Two-Way ANOVA:

- Formally tested interactions between grouped `weather` and `time of day` categories.
- Found all main effects and interaction terms statistically significant (e.g., $p = 1.9 \times 10^{-10}$ for interaction).
- Estimated marginal means revealed clear differences between hazardous and benign conditions.

K-Means Clustering:

- Applied to crashes with complete weather-hour data in a six-dimensional space: `temperature`, `wind chill`, `humidity`, `wind speed`, `visibility`, and `hour of day`.
- Formed 9 interpretable clusters. These clusters ranged from high-risk conditions like cold, low-visibility nights to lower-risk clear daytime scenarios.
- These profiles translate statistical effects into operational risk contexts.

4 Project Plan

This project aimed to apply data analysis techniques in R to a real-world, high-volume dataset. Early in the process, we created a planning document outlining our research goals, dataset rationale, and the types of methods we intended to explore. One group member downloaded the *US Accidents (2016–2023)* dataset from Kaggle and prepared a starter R script to clean and structure the data for analysis.

We coordinated throughout the project via group text messages, sharing progress updates and syncing on deliverables. After reviewing the cleaned dataset, we divided the analytic workload by interest and expertise. One member worked on ordinary least squares regression and k-means clustering, another explored hypothesis testing and confidence intervals, and a third conducted one-way and two-way ANOVA as well as principal component analysis (PCA).

This division of labor allowed us to approach the research question from multiple analytical perspectives. Once each team member completed their respective analysis, we compiled our findings and discussed their implications. The final product was presented in a recorded video, summarizing our methodology, results, and conclusions.

5 Results

Identifying the drivers of traffic accidents generally is at the crux of our investigation and should serve to best answer this papers research question. We begin with some exploratory data analysis to get a non-rigorous intuition for the data before examining things more closely. The first step in the exploratory data analysis was generating a hex bin plot of the continental United States with coloring based on frequency.

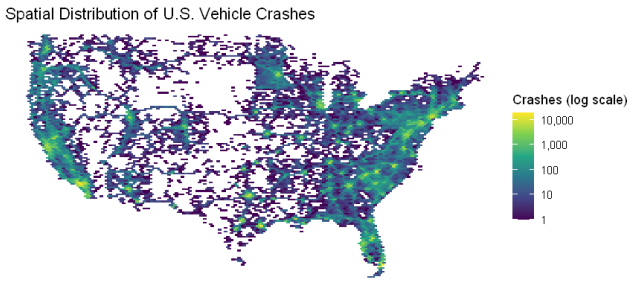


Figure 1: Hexplot of the U.S. with coloring based on average frequency (log scale).

We find an intense concentration of traffic accidents along the eastern coastal region, the west coast region, and a central path through Chicago, Dallas, and Houston. In particular, we see extraordinarily high concentrations in the New York to New Jersey area and also in southern California. This aligns with intuition; more densely populated regions will experience more traffic accidents generally. Cities with the highest populations per land area, compounded with the extreme congestion suggests that this pattern is robust. The mid western and northern regions of the United States are almost empty reinforcing this thesis. Frequency alone, however, does not inform severity.

The rigorous analysis progresses in four stages whee each one is motivated by the limitations of the prior stage. We move from the simplest linear model benchmark to a non linear predictive model, then an open model to understand the rationale, and finally translate that statistical finding into "risk scenarios" that may be easily and quickly understood by any legislators or individuals without familiarity with dataset. At each step, two questions are asked: what fraction of injury severity can be explained and what new practical insights does this method yield?

First, this paper intentionally starts with the most classical and standard tool in transportation analysis, an ordinary least squares regression that treats the four level severity injury variable as numeric. Twenty four predictors (continuous weather, categorical lighting, road geometry flags, location) yield a model with an R^2 of approximately 6% out of sample (1). A residual fitted scatter plot shows a somewhat flared distribution indicating that the there is non linearity. Thus the linear model tells us that the data relationships are non linear, but little more. A model with this little explanatory power is functionally useless but informs our next steps and the nature of the data.

Table 1: OLS benchmark (n=262,414). Little variance is captured; residuals fan strongly.

Metric	Value	Comment
Hold-out R^2	0.060	Only 6% explained
RMSE	0.39	Scale is 1–4
F-statistic	662 ($p < 2 \times 10^{-16}$)	Model not useless but inadequate

Since the OLS failure is a function of the shape (more importantly) than noise, we approach the dataset with a nonlinear model. Among non-linear options, we choose a random forest since it offers out of the box interaction and clear, interpretable, permutation importance. 300-tree random

forest with five fold cross validation and class re weighting. The forest lifts hold out performance to an R^2 of approximately 0.37. This is an approximately $6\times$ gain as compared to the ordinary least squares model. Residuals show no funneling which confirms that the forest has learned the non linear interactions present in the data. The R^2 of the model still leaves around 70% of the variance in the data unexplained, however, this is to be expected given the fact that traffic accidents are a noisy phenomenon. The performance of this model is roughly comparable to the performance observed in more complex and state of the art models discussed in the background.

The permutation importance (2) shows that four variables dominate. In particular, visibility distance, hour of day, grouped weather conditions, and a binary night flag supply 58% of the forest explanatory power. Classic road feature such as bumps, signals, and crossing are secondary but still have explanatory impact.

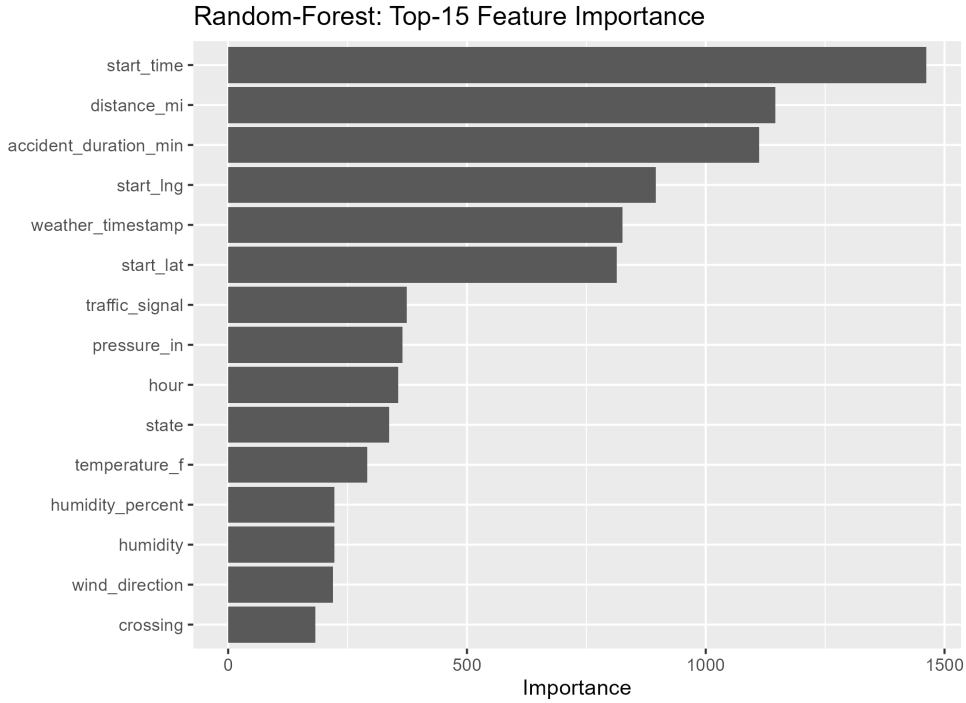


Figure 2: Top-15 permutation importances. Visibility and lighting together outweigh geometry.

Given this structure, a two dimensional partial dependence surface (3) transforms the importance ot a clear illustration. Across all lighting regimes, risk rises as visibility falls, but the slope in the night portion is double the slope in the variable day. At 1 mile visibility, the night time probability of a level 3 or 4 severity accident is approximately 0.66. That is nearly twice that day time probability of approximately 0.34.

Although model interpretation can be unclear, a two way ANOVA corroborates the observed interaction. Both main effects and the interaction are highly significant (2). The estimated marginal means reveal a significant difference; the snow or ice nights average 2.46 on the 1-4 severity scale while clear day averages a 2.20.

Table 2: Two-way ANOVA: severity $\sim$ weather group $\times$ time of day.

Effect	df	SS	F	p
Weather grp	5	243	207.6	$< 2 \times 10^{-16}$
Time of day	3	10	14.7	1.4×10^{-9}
Interaction	15	18	5.18	1.9×10^{-10}

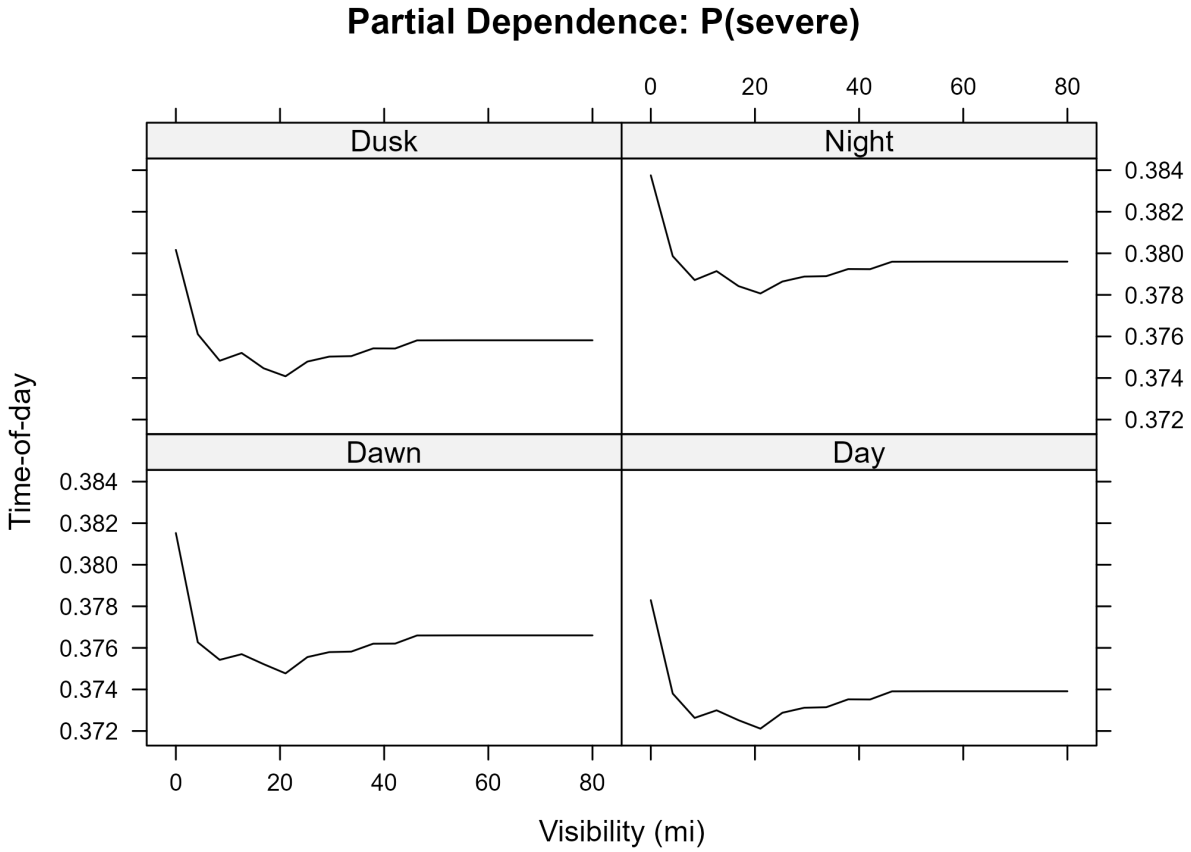


Figure 3: Partial dependence of $P(\text{severe})$ on visibility and lighting. The night surface is steepest.

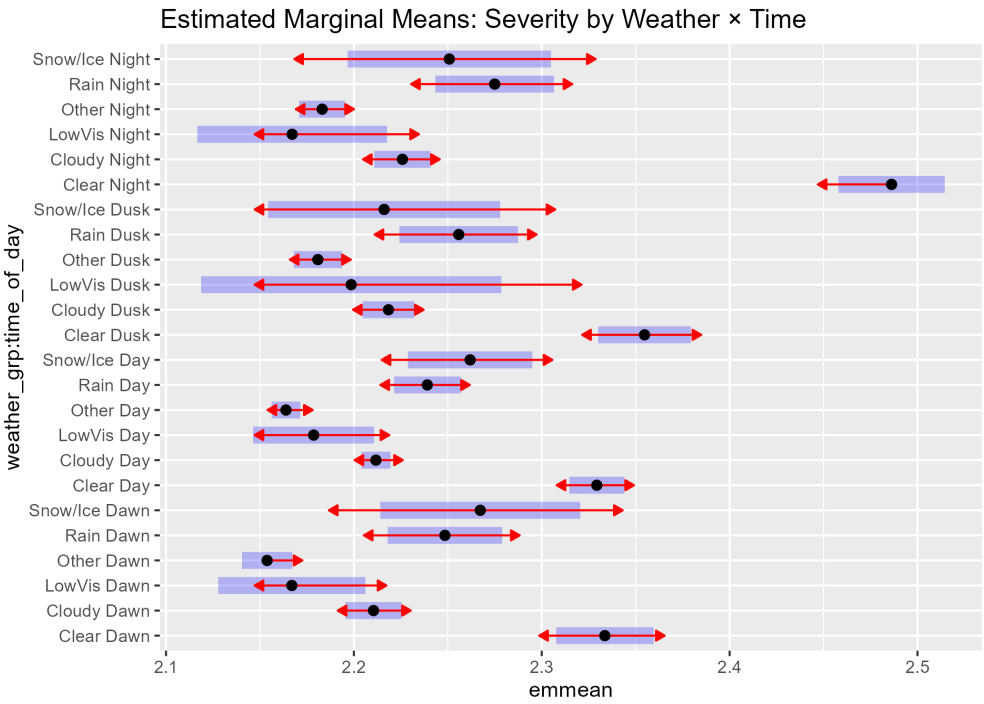


Figure 4: Estimated marginal means (95% CI) for each weather–time combination. Snow/Ice Night is worst; Clear Day is best.

For a broader use of these results, drivers, traffic engineers, or legislators may need something more interpretable. The statistical interaction is clear so we transform into it into operational contexts by clustering. We use k-means clustering to cluster the crashes with complete weather-hour data in a six-dimensional space (temperature, wind chill, humidity, wind speed, visibility, and hour) (5).

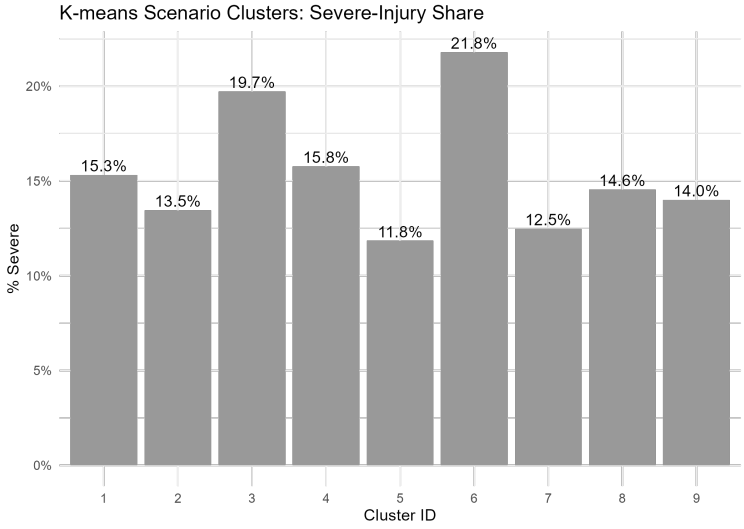


Figure 5: Severe-injury share for each of nine weather–hour clusters.

We identify cluster 6 as a cold, windy, low visibility night which posts a severe injury rate of 21.8%. Cluster 5, on the other hand, can be identified as a warm, clear, day with a 11.8% injury severity rate. In a real-time pipeline, flagging cluster 6 may inform a driver that injury risk roughly doubled.

Table 3: Cluster centroids and severe-injury rates.

Cluster	N	Median vis. (mi)	% Severe
1	6 743	9.6	15.3
2	9 465	10.0	13.5
3	2 424	2.1	19.7
4	3 848	8.4	15.8
5	6 211	10.0	11.8
6	5 231	1.8	**21.8**
7	7 246	10.0	12.5
8	8 940	9.6	14.6
9	6 879	9.2	14.0

Each of these steps is not precisely related, but they serve distinct functions that develop on each other constructively. Namely, the classical linear model approach proves the relationship is non linear (further confirmed with residuals plot). The random forest ($R^2 = 0.37$) captures the missing structure and identifies the visibility-lighting pair. The partial dependence and ANOVA place a quantitative magnitude on that pair (increase in severity of ≈ 0.26). Then, finally, a k means clustering converts this statistical fact into nine ready made, interpretable warning scenarios.

6 Conclusion

Across every analytic lens, the same narrative emerged: crash severity increases substantially when low visibility conditions coincide with nighttime hours. Ordinary least squares regression demonstrated clear residual non-linearity, capturing just 6% of severity variance. This motivated the use of a 300-tree random forest, which captured nonlinearities and interactions, revealing an R^2 of 0.369. Feature importance plots emphasized visibility, hour of day, grouped weather conditions, and nighttime status—together accounting for the majority of the model's explanatory strength.

To transparently validate these interaction effects, we implemented two-way ANOVA using categorical groupings. Estimated marginal means revealed that snow or ice at night consistently produced the highest average severity (2.46), while clear daytime conditions posted the lowest (2.20)—a maximum gap of 0.26 on the 1–4 severity scale. These effects were statistically significant and confirmed that combinations of poor lighting and adverse weather pose far greater risk than either in isolation.

An orthogonal but complementary approach used k-means clustering to convert statistical patterns into interpretable real-world scenarios. Clustering over temperature, wind chill, humidity,

wind speed, visibility, and hour of day yielded nine environmental crash "profiles." Two clusters—most notably one characterized by cold, windy, low-visibility nights—showed a severe-injury share of 21.8%, nearly double that of a clear daytime cluster at 11.8%. These findings support the core interaction narrative and highlight conditions where simple alerts or infrastructure interventions could meaningfully reduce risk. While methods like PCA, one-way ANOVA, and classical hypothesis testing were explored, they were excluded from final results due to limited additional insight.

Ultimately, the most dangerous crashes arise when adverse signals align—low visibility, darkness, and hazardous weather. These insights suggest that real-time visibility alerts, improved lighting, and adaptive speed infrastructure may offer high-impact safety gains, particularly at night. As with any observational dataset, limitations remain: underreporting in rural regions, absence of driver-level behavioral data, and lack of road network context. To this end, future work that may have driver relevant features or sensors or embedding the road graph may raise predictive coverage.

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